# 12th hacker series

# Dual nature of radiation and matter

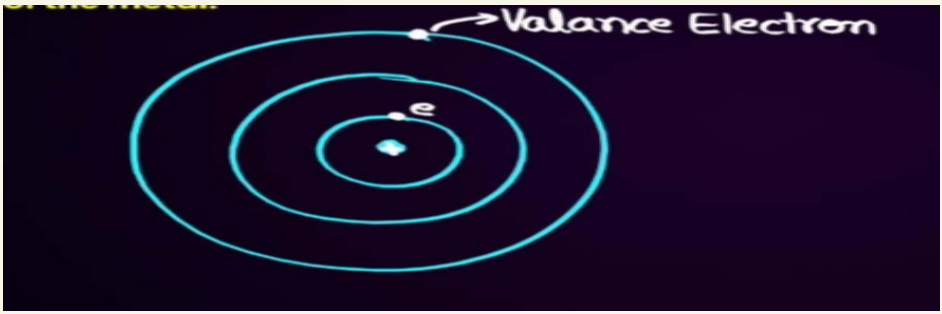
## Free electrons in metals

- the minimum amount of energy required by an electron to just escape from the metal surface is called work function of the metal

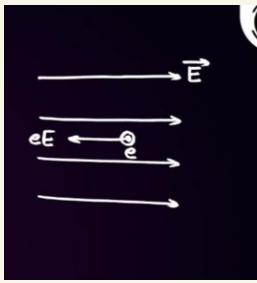
$W = h\nu_0$

→ Threshold Frequency

h → plank's constant = $6.6 \times 10^{-34}$ J.s.

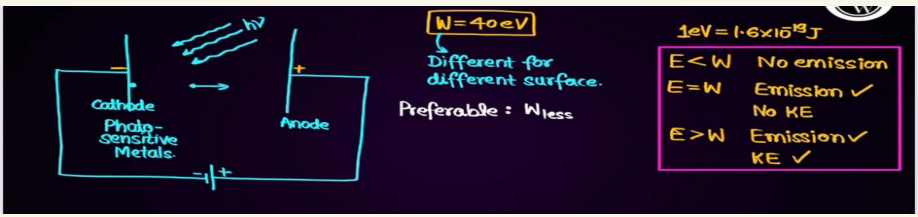



## Electron emission



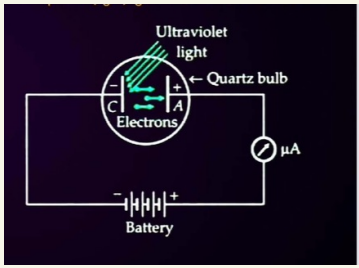

- thermionic emission- heat
- Field emission, or cold cathode emission- electric field
- Photoelectric emission- light
- Secondary emission- particles

## Photoelectric emission



# Photoelectric effect

- the phenomena of emission of electrons from a metal surface when electromagnetic radiation of sufficiently high frequency are incident on it is called photoelectric effect. The photo generated electrons are called photo electrons.
- Alkali metals like lithium ,sodium ,potassium ,Cs and Rb are highly photosensitive. They emit electrons even with visible light.
- metals like zinc, magnesium, aluminium, et cetera, respond only to ultraviolent light
- X-rays can eject electrons even from heavy metals




**QUESTION**

When ultraviolet rays are incident on a metal plate, then the photoelectric effect does not occurs. It occurs by the incidence of

OD 2004

(a) X-rays
(b) radio wave
(c) infrared rays
(d) green house effect

**QUESTION**

Which one of the following metals does not exhibit emission of electrons from its surface when irradiated by visible light?

OD 2023

(a) Rubidium
(b) Sodium
(c) Cadmium
(d) Caesium

**QUESTION**

Which one of the following metals does not exhibit emission of electrons from its surface when irradiated by visible light?

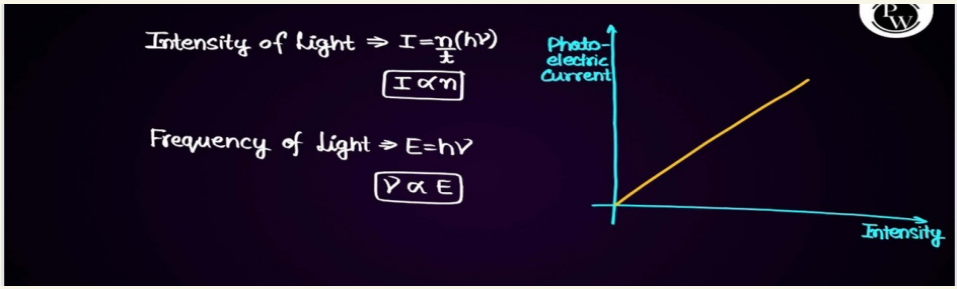
OD 2023

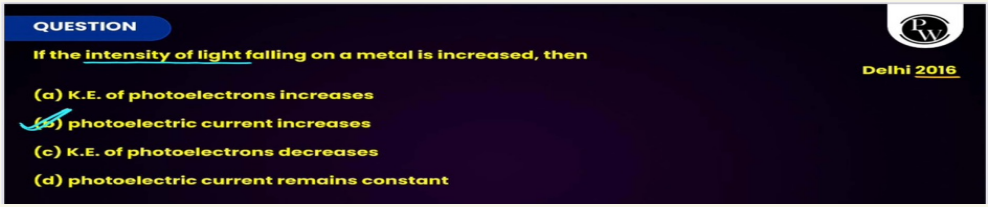
(a) Rubidium
(b) Sodium
(c) Cadmium
(d) Caesium

Certain metals like zinc, cadmium, magnesium etc. respond only to UV light to cause electron emission from its surface.
Thus (c) is correct option.

# Experimental study of photoelectric effect

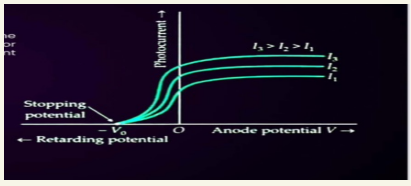
- effect of intensity of light and photoelectric current
  - since the photoelectric current is directly proportional to the number of photo electrons, emitted per second. This implies that the number of photo electrons emitted per second is proportional to the intensity of incident radiation.

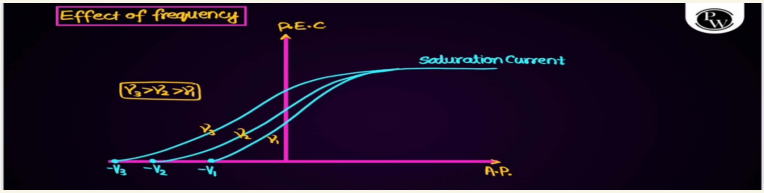


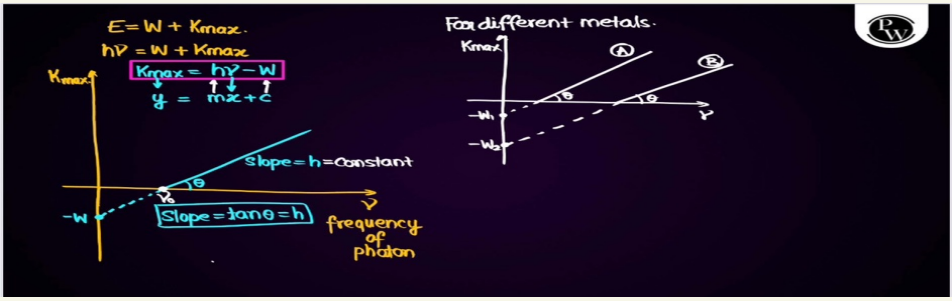

- effect of potential
  - the photoelectric current increases with the increase in accelerating potential till a stage is reached when the photoelectric current becomes maximum and does not increase further with the increase in the accelerating potential. This maximum value of the photoelectric current is called the saturation current.
  - The value of the retarding potential at which the photoelectric current becomes zero is called cut-off or stopping potential for the given frequency of the incident radiation

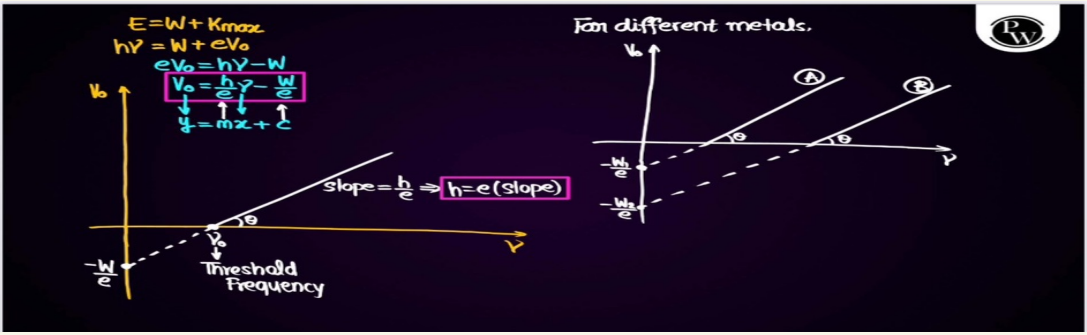
$$K_{max} = \frac{1}{2}mv^2_{max} = eV_0$$





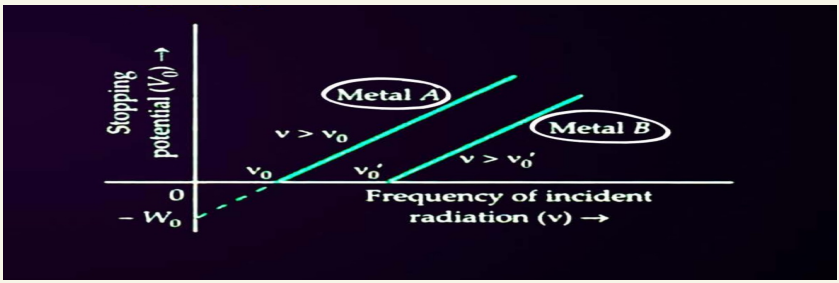

- effect of frequency of incident radiation on stopping potential
- the value of stopping potential increases with the frequency of incident radiation. for frequencies $\nu_3 > \nu_2 > \nu_1$ The corresponding stopping potential vary in the order $\nu_3 > \nu_2 > \nu_1$






- the stopping potential increases nearly with the frequency of the incident radiation for a given photosensitive material
- for two different metals, A and B these graphs are parallel straight lines that is, they are same slope, but the threshold frequencies are different for the two metals




## Law of photoelectric emission

- for a given photo, sensitive material and frequency of incident radiation (above the threshold frequency) the photoelectric current is directly proportional to the intensity of light. The saturation current is directly proportional to the intensity of incident radiation.
- For a given photo sensitive material. There exist a certain minimum cut-off frequency below which no electrons are emitted. However, high is the intensity of incident radiation. This frequency is called threshold frequency.
- Above the threshold frequency, the stopping potential or equivalent lead. The maximum kinetic energy of the photo electrons is directly proportional to the frequency of incident radiation, but is independent of its intensity.
- The photoelectric emission is an instantaneous process. The time lag between the incident of light, radiation and emission of electron is very small, even less than $10^{-9}$s

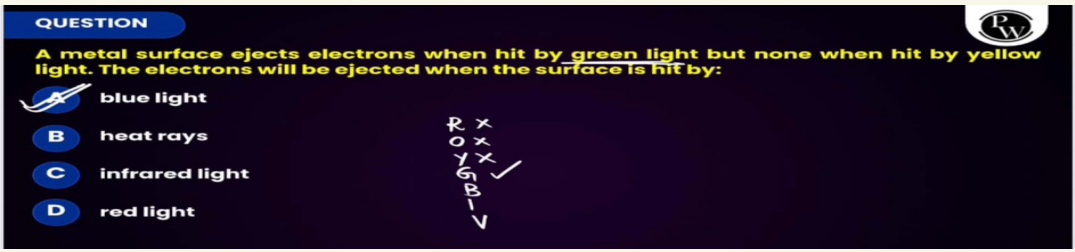
## Failure of classical wave theory

- according to the wave theory, when a wavefront of light strikes a metal surface, the free electrons and the surface absorb the radiant energy continuously, greater the intensity of incident radiation, greater are the amplitude of electric and magnetic fields

And greater is the energy density of the wave. Hence, higher intensity should liberate photo electrons with greater kinetic energy. But this is contrary to the experimental result that the maximum kinetic energy of the photo electrons does not depend on the intensity of the incident radiation.

- no matter what the frequency of incident radiation is a light wave of sufficient intensity should be able to impart enough energy required to eject the electrons from the metal surface that the wave theory fails to explain the existence of the threshold frequency
- The energy of light wave is smoothly and evenly distributed across its advancing wavefront, each electron interception in significantly small amount of this energy, and so it should require a finite time to escape from the metal surface, but actually, the emission is almost instant instantaneous






**QUESTION**

In photoelectric effect, the photocurrent:

A. depends both on intensity and frequency of the incident light.

B. does not depend on the frequency of incident light but depends on the intensity of the incident light.

C. decreases with increase in frequency of incident light.

D. increases with increase in frequency of incident light.

**QUESTION**

The maximum value of photoelectric current is called:

A. base current

B. saturation current

C. collector current

D. emitter current

**QUESTION**

The monochromatic beams $A$ and $B$ of equal intensities I, hit a screen. The number of photons hit on the screen by beam $A$ is twice that by beam B. The ratio of their frequencies will be:

A. 1 : 2

B. 2 : 1

C. 1 : 1

D. 1 : 3

$I_A = I_B$

$n_A = 2n_B$

$n_A(h\nu_A) = n_B(h\nu_B)$

$2n_B\nu_A = n_B\nu_B$

$\frac{\nu_A}{\nu_B} = \frac{1}{2}$

**QUESTION**

The energy of emitted photo electron depends upon

Delhi 2001, SQP 2014

$\lambda \uparrow \nu \downarrow$

$\lambda \downarrow \nu \uparrow$

(a) Intensity of light

(b) Wavelength of light

(c) Work-function of metal

(d) None of these

$E = W + K_{max}$

$h\nu = W + K_{max}$

$\frac{hc}{\lambda} = W + K_{max}$

$W$ → Constant

**QUESTION**

There are $n_1$ photons of frequency $\nu_1$ in a beam of light. In an equally energetic beam, there are $n_2$ photons of frequency $\nu_2$. Their correct relation is

OD 2009, Delhi 2012

$n_1(h\nu_1) = n_2(h\nu_2)$

$\frac{n_1}{n_2} = \frac{\nu_2}{\nu_1}$

(a) $\frac{n_1}{n_2} = 1$

(b) $\frac{n_1}{n_2} = \frac{\nu_1}{\nu_2}$

(c) $\frac{n_1}{n_2} = \frac{\nu_2}{\nu_1}$

(d) $\frac{n_1}{n_2} = \frac{\nu_1^2}{\nu_2^2}$

QUESTION

Assertion : Photo-sensitivity of a metal is high if its work function is small.
Reason : Work function $= hf_0$ where $f_0$ is the threshold frequency.

(a) Both Assertion and Reason are correct and Reason is a correct explanation of Assertion.
(b) Both Assertion and Reason are correct but Reason is not a correct explanation of Assertion.
(c) The Assertion is correct but Reason is incorrect.
(d) Both the Assertion and Reason are incorrect.

## Einstein's theory of photoelectric effect

The main points of the Einstein's theory of photoelectric effect are

- photoelectric animation is the result of interaction of two particles- 1, a photon of incident radiation and other an electron of photo sensitive metal
- The free electrons are bound within the metal due to restraining forces on the surface. The minimum energy required to liberate an electron from the metal surface is called work function $W_0$ of the metal
- Each photon interacts with one electron. The energy $h\nu$ of the incident photon is used up in two parts
- (a) A part of the energy of the photon is used in liberating the electron from the metal surface, which is equal to the work function $W_0$ of the metal and
- (b) the remaining energy of the photon is used in imparting kinetic energy to the ejected electron
- Very few (less than 1%) Photons, whose energies are greater than $W_0$ are capable of ejecting the photoelectrons
- by the conservation of energy, energy of the incident photon = maximum kinetic energy of photoelectron + work function

$$h\nu = \frac{1}{2}mv_{max}^2 + W_0$$

QUESTION

What is the frequency of a photon whose energy is 66.3 eV?

A $1.6 \times 10^{18}$ Hz
B $3.2 \times 10^{16}$ Hz
C $1.6 \times 10^{16}$ Hz
D $3.2 \times 10^{18}$ Hz

$h\nu = 66.3\text{eV}$

$\nu = \frac{66.3 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}}$

$= 1.6 \times 10^{16}$

**QUESTION** (1mb)

**Monochromatic light of frequency $6.0 \times 10^{14}$ Hz is produced by a laser. The power emitted is $2.0 \times 10^{-3}$ W. (i) What is the energy of each photon in the light? (ii) How many photons per second, on the average, are emitted by the source?** [NCERT]

(A) $3.98 \times 10^{-19}$ J and $6.0 \times 10^{15}$

(B) $3.98 \times 10^{-19}$ J and $5.0 \times 10^{15}$

(C) $4.98 \times 10^{-19}$ J and $5.0 \times 10^{15}$

(D) $4.98 \times 10^{-19}$ J and $6.0 \times 10^{15}$

(ii) $P = \frac{E}{t} = \frac{n(h\nu)}{t}$

$2 \times 10^{-3} = \frac{n}{t}(3.96 \times 10^{-19})$

$\frac{n}{t} = \frac{2 \times 10^{-3}}{3.96 \times 10^{-19}}$

$= 0.5 \times 10^{16}$

$= 5 \times 10^{15}$

(i) $E = h\nu$

$= 6.6 \times 10^{-34} \times 6 \times 10^{14}$

$= 39.6 \times 10^{-20}$

$= 3.96 \times 10^{-19}$ J.

**QUESTION**

**The number of Photons of frequency $10^{14} Hz$ in radiation of 6.62 J will be**

CBSE 2018

(a) $10^{10}$

(b) $10^{15}$

(c) $10^{20}$

(d) $10^{25}$

$E = n(h\nu)$

$6.62 = n(6.6 \times 10^{-34} \times 10^{14})$

$n = 10^{20}$

**QUESTION** (3 Marks)

**The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 eV.** [NCERT]

$W = 2.14$ eV.

(a) $h\nu_0 = 2.14$ eV

$\nu_0 = \frac{2.14 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}}$

$= 0.519 \times 10^{15}$

$\nu_0 = 5.19 \times 10^{14}$ Hz

(b) $E = W + K_{max}$

$\frac{hc}{\lambda} = 2.14 \times 1.6 \times 10^{-19} + 1.6 \times 10^{-19}(0.6)$

$\frac{hc}{\lambda} = 2.74 \times 1.6 \times 10^{-19}$

$\lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{2.74 \times 1.6 \times 10^{-19}}$

$\lambda = 4.5 \times 10^{-7}$ m.

**QUESTION**

**If 5% of the energy supplied to an incandescent light bulb is radiated as visible light, how many visible light photons are emitted by 100 watt bulb? Assume wavelength of all visible photons to be 5600 Å. Given $h = 6.625 \times 10^{-34}$ Js.**

**Solution:** Here $\lambda = 5600\text{Å} = 5600 \times 10^{-10} m$

Energy of one photon

$$= \frac{hc}{\lambda} = \frac{6.625 \times 10^{-34} \times 3 \times 10^{8}}{5600 \times 10^{-10}} = 3.5 \times 10^{-19} J$$

A $100 W$ bulb supplies $100 J$ of energy per second.

$\therefore$ Energy released per second as visible photons

$$= \frac{100 \times 5}{100} = 5 J$$

$\therefore$ Number of photons emitted per second as visible light

$$= \frac{5}{3.5 \times 10^{-19}} = 1.43 \times 10^{19}.$$

## Dual nature of matter: De broglie waves

- the waves associated with material particles in motion are called matter or de broglie waves, and their wavelength is called de broglie wavelength
- De broglie wave equation:
- Considering photon as a particle of mass m, the energy associated with it is given by Einstein's mass energy relationship as

$$E = h\nu = mc^2$$

$$\frac{hc}{\lambda} = mc^2$$

$$\boxed{\frac{h}{\lambda} = p}$$

$$\boxed{\lambda = \frac{h}{p}} \rightarrow \text{de-broglie wavelength}$$

$$p = \sqrt{2mK}$$

$$\boxed{\lambda = \frac{h}{\sqrt{2mK}}}$$

for electrons,

$$\boxed{\lambda_e = \frac{h}{\sqrt{2m(eV)}}}$$

$$= \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} V}}$$

$$\boxed{\lambda_e = \sqrt{\frac{150}{V}}\ \text{Å}}$$

$$\boxed{\lambda_e = \frac{12.3}{\sqrt{V}}\ \text{Å}}$$

If a radiation of energy $E$ falls normally on a perfectly reflecting surface, then the momentum transferred to the surface is

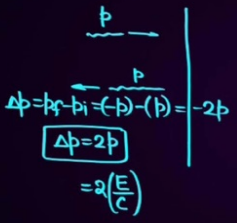
Delhi 2015

(a) $E_c$

(b) $Ec^2$

(c) $\frac{E}{c^2}$

(d) $\frac{2E}{c}$




$E=mc^2$

$\frac{E}{c}=mc=p$

**QUESTION**

If momentum of a particle is doubled, then its de-Broglie's wavelength will

SQP 2010

(a) be half

(b) be two times

(c) be four times

(d) remain unchanged

$\lambda=\frac{h}{p}$

$\lambda'=\frac{h}{2p}=\frac{\lambda}{2}$

**QUESTION**

A particle of mass $m$ and charged $q$ is accelerated through a potential $V$. The De-Broglie wavelength of the particle will be:

SQP 2018, Comp 2004

(a) $\frac{Vh}{\sqrt{2qm}}$

(b) $\frac{q}{\sqrt{2mV}}$

(c) $\frac{h}{\sqrt{2qmV}}$

(d) $\frac{mh}{\sqrt{2qV}}$

$\lambda=\frac{h}{p}=\frac{h}{\sqrt{2m(qV)}}$

**QUESTION**

What is the de Broglie wavelength associated with (i) an electron moving with a speed of $5.4 \times 10^6 \text{ms}^{-1}$ and (ii) a ball of mass $1.50\text{g}$ travelling at $30.0\text{m/s}$? [NCERT]

**Solution:** (i)

$$\lambda_e = \frac{h}{m_e v_e} = \frac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 5.4 \times 10^6}$$
$$= 0.135 \times 10^{-9} m = 0.135 nm.$$

(ii)

$$\lambda_b = \frac{h}{m_b v_b} = \frac{6.63 \times 10^{-34}}{0.150 \times 30.0} = 1.47 \times 10^{-34} m.$$

**QUESTION**

**What is the de Broglie wavelength associated with an electron, accelerated through a potential difference of 100 volts?** [NCERT]

**Solution:** Accelerating potential, $V = 100V$
The de Broglie wavelength is

$$\lambda = \frac{1.227}{\sqrt{V}}\,nm = \frac{1.227}{\sqrt{100}}\,nm \simeq 0.123nm.$$

This wavelength is of the order of X-ray wavelengths.

$\lambda_e = \frac{12.3}{\sqrt{V}}\,\text{Å}$

$= \frac{12.3}{\sqrt{100}}\,\text{Å}$

$\lambda_e = 1.23\,\text{Å}$

**QUESTION**

**Find the typical de-Broglie wavelength of an electron in a metal at 27°C and compare it with the mean separation between two electrons in a metal which is given to be about $2 \times 10^{-10}$ m** [NCERT]

A $6.2 \times 10^{-9}$ m

B $5.2 \times 10^{-9}$ m

C $4.2 \times 10^{-9}$ m

D $3.2 \times 10^{-9}$ m

$\lambda_e = \frac{h}{\sqrt{2mK}}$

$= \frac{h}{\sqrt{2m\left(\frac{3}{2}kT\right)}}$

$= \frac{h}{\sqrt{3mkT}}$

$= \frac{6.6 \times 10^{-34}}{\sqrt{3 \times 9.1 \times 10^{-31} \times 1.38 \times 10^{-23} \times 300}}$

$= 6.2 \times 10^{-9}$ m.

$E = \frac{3}{2}kT$

↓

Boltzmann's Constant

$1.38 \times 10^{-23}$

$T = 27 + 273 = 300$ K

$\frac{\lambda_e}{2 \times 10^{-10}} = 31$

**QUESTION**

**A particle is dropped from a height *H*. The de Broglie wavelength of the particle as a function of height is proportional to** [NCERT Exemplar]

A $H$

B $H^{1/2}$

C $H^0$

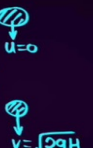
D $H^{-1/2}$

$\lambda = \frac{h}{p}$

$= \frac{h}{mv}$

$\lambda = \frac{h}{m\sqrt{2gH}}$

$\lambda \propto \frac{1}{\sqrt{H}}$




**QUESTION**

**Consider a beam of electrons (each electron with energy $E_0$) incident on a metal surface kept in an evacuated chamber. Then** [NCERT Exemplar]

A no electrons will be emitted as only photons can emit electrons.

B electrons can be emitted but all with an energy, $E_0$.

C electrons can be emitted with any energy, with a maximum of $E_0 - \phi$ ($\phi$ is the work function).

D electrons can be emitted with any energy, with a maximum of $E_0$.

**QUESTION**

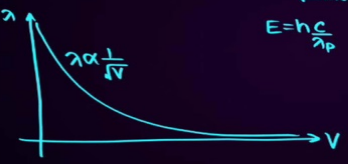
**Show graphically the variation of the de-Broglie wavelength ($\lambda$) with the potential ( $V$ ) through which an electron is accelerated from rest.**

**OD 2020, Comp 2018**

$\lambda = \frac{h}{\sqrt{2mqV}}$

$\lambda_e = \frac{h}{\sqrt{2mk}}$

$E = h\frac{c}{\lambda_p}$




1 **(a) Electron emission:** The phenomenon of emission of electrons from the surface of metal by supplying energy is called electron emission.

**(b) Types of electron emission:**

(i) **Thermionic emission:** The emission of electrons from the surface of metal by heating is called thermionic emission.

(ii) **Photoelectric emission:** The emission of electrons from the surface of metal, when light of suitable frequency is incident on metal surface is called photoelectric emission.

(iii) **Field emission:** The emission of electrons from surface, when a strong electric field is applied across the metal surface is called field emission.

2. **(a) Work function ($\phi_0$):** The minimum amount of energy required to remove free electrons from the metal surface is called work function. Unit of work function ($\phi_0$) is e V.

(b) 1 eV : The energy is said to be 1 eV when an electron is accelerating through a potential difference of 1 volt.

$1eV = 1.6 \times 10^{-19}$ J

**Note:** (a) The work function of a metal depends upon: (i) the properties of the metal, (ii) nature of the surface of metal.

(b) Work function, $\phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$, where $\nu_0$ and $\lambda_0$ are the threshold frequency and threshold wavelength, respectively.

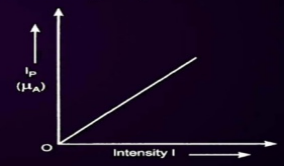
3 **Photoelectric effect:** The phenomenon of emission of electrons from the surface of metal, when a light beam of suitable frequency is incident on the metal surface, is called photoelectric effect. The electrons emitted during this process are called photoelectrons and current so produced is called photoelectric current.

4 **Threshold frequency:** The minimum frequency of incident light required to emit electrons from a metal surface is called threshold frequency.

**Note:** If the incident frequency of the light is less than the threshold frequency of the metal, then photoelectric effect is not possible.

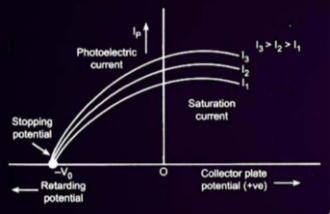
5 **Experimental study of photoelectric effect:**

**(a) Effect of intensity of incident light on photoelectric current:** Photoelectric current (number of photoelectrons per second) is directly proportional to the intensity of incident light, provided the incident frequency is greater than the threshold frequency.




**(b) Effect of potential difference on photoelectric current:**

For a given frequency and intensity of incident radiation, the photoelectric current increases with increase in the potential applied to the collector.




**Observation from graph:**

(i) When all the emitted electrons reach the collector plate (anode) and if we further increase the potential applied to anode plate, then photoelectric current does not increase, it attain its maximum value, called saturation current.

(ii) Cut-off potential: The minimum negative potential given to the anode for which photoelectric current becomes zero called cutoff potential (Retarding potential or stopping potential).

$(K.E.)_{max} = e\,V_0$

or $\frac{1}{2}mv_{max}^2 = e\,V_0$

**Note:**

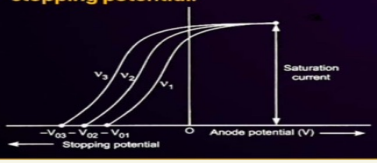
(i) Saturation of photoelectric current depends upon the intensity of incident light.

(ii) The stopping potential does not depends upon the intensity of incident light.

(iii) Maximum kinetic energy of emitted photoelectrons does not depends upon intensity of incident light.

(iv) Emitted photoelectrons may have different energies.

**(c) Effect of frequency of incident light on stopping potential:**




At a constant intensity of light, but the frequency of intensity is changed, we find that the value of stopping potential depends upon the frequency of the incident light provided frequency is greater than the threshold frequency.

It means that the stopping potential is directly proportional to the frequency of incident light.

$$V_0 \propto \nu$$

**6 Laws of photoelectric emission:**

(i) For a given material and frequency of incident light, the photoelectrons ejected per second (photoelectric current) is directly proportional to the intensity of the incident light.

(ii) For a given material, there is a threshold frequency below which no photoelectric emission takes place.

(iii) For a given material, the maximum kinetic energy of photoelectrons increases with increase in the frequency of incident light, provided the frequency of incident light is greater than the threshold frequency.

(iv) The process of photoelectric emission is instantaneous.

**7. Einstein photoelectric equation:** When a photon of energy $(E = h\nu)$ falls on metal surface, this energy of photon is used (i) to overcome the work function and (ii) rest part is used to give kinetic energy to the emitted electrons.

$$\boxed{h\nu = \phi_0 + (K.E.)_{max}}$$

$$(K.E.)_{max} = \frac{1}{2}mv^2_{max} \rightarrow eV_0$$

**Note:**

(a) According to Planck's quantum theory, light radiation consists of tiny packets of energy called quanta. One quantum of light radiation is called a photon which travels with a speed of light.

(b) Einstein's photoelectric equation establishes the particle nature of light.

(c) Einstein's photoelectric equation in terms of wavelength of incident light

$$\frac{1}{2}mv^2_{max} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

$$\text{i.e., } (K.E.)_{max} = \frac{hc(\lambda_0 - \lambda)}{\lambda\lambda_0}$$

(d) Wavelength of incident light must be less than the threshold wavelength for the photosensitive material.

**8. Relation between stopping potential $(V_0)$ and threshold frequency $(\nu_0)$:**

$$(K.E)_{max} = h\nu - \phi_0$$

$$(K.E.)_{max} = h(\nu - \nu_0) \quad (\phi_0 = h\nu_0)$$

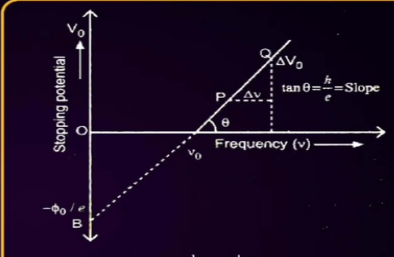
$$eV_0 = h(\nu - \nu_0) \quad [(K.E.)_{max} = eV_0]$$

$$\therefore V_0 = \frac{h}{e}(\nu - \nu_0)$$

$$V_0 = \frac{hc}{e}\left[\frac{1}{\lambda} - \frac{1}{\lambda_0}\right]$$

$$\left(\nu = \frac{c}{\lambda}, \nu_0 = \frac{c}{\lambda_0}\right)$$

For photoelectric effect: $\lambda < \lambda_0$ and $\nu > \nu_0$




$$\Rightarrow \quad V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

$$V_0 \propto \nu$$

$$\frac{h}{e} = \tan\theta = \frac{\Delta V_0}{\Delta\nu}$$

$\therefore$ $h = e \times$ Slope of $V_0$ versus $\nu$ graph

Intercept $(OB) = -\phi_0/e$

$\therefore$ $\phi_0 = -($ Intercept $OB) \times e$

**9. (a) Particle nature of light-The photon:**

Photon is a packet of energy or quantum of energy emitted at the speed of light by an emitter.

$E = h\nu$ Photoelectric effect shows that light consists of packets, For $n$ photons

$E = nh\nu$ so photon confirm the particle nature of light.

**(b) Properties of a photon:**

(i) Photon moves with a speed of light.
(ii) Rest mass of a photon is zero.
(iii) Photon is electrically neutral.
(iii) Energy of a photon is,

$$E = \frac{hc}{\lambda} = \frac{1240\text{eV}}{\lambda}$$

(iv) Photon travel in a straight line.

**10. Dual nature of light:** The phenomenon like interference, diffraction and polarisation are explained on the basis of wave theory of electromagnetic radiation. On the other hand, photoelectric effect shows that light has a particle nature. Hence, we assume dual nature of light.

**11. Wave nature of particles (de-Broglie Hypothesis):** A wave is associated with moving material particle is called matter wave or deBroglie wavelength. de-Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2meV}} = \frac{12.27\text{Å}}{\sqrt{V}}$$

where,
h = Planck's constant
v = Velocity of moving particle
K = Kinetic energy
V = Potential
m = Mass of electron

## Dual Nature of Radiation & Matter

**Work Function**
$W_0 = h\nu_0 = \frac{hc}{\lambda_0}$

**Momentum of Photon**
$p = \frac{h\nu}{c} = \frac{h}{\lambda}$

**Rest Mass of Photon**
$m = 0$

**Kinetic Mass of Photon**
$m = \frac{h\nu}{c^2} = \frac{h}{c\lambda}$

**Einstein's Equation**
$h\nu = h\nu_0 + K_{max}$

**Max. Kinetic Energy**
$K_{max} = \frac{1}{2}mv_{max}^2 = eV_0$

**Energy of Photon**
$E = h\nu = \frac{hc}{\lambda} = \frac{1240}{\lambda}\text{eV}$

**de-Broglie Wavelength**
$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}$

**de-Broglie λ of electron.**
$\lambda_e = \frac{h}{\sqrt{2meV}} = \sqrt{\frac{150}{V}}\,A°$
$\lambda_e = \frac{12.27}{\sqrt{V}}\,A°$

$\lambda = \frac{h}{\sqrt{2mkT}}$